made on these territories, we acknowledge the harms and mistakes of the past, and we dedicate ourselv
move forward in partnership with Indigenous communities in a spirit of Reconciliation and collaboration.

Principal component analysis

MATH 2740 – Mathematics of Data Science – Unit U05

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Outline

Dimensionality reduction

In the previous unit, we saw that SVD could be used to reduce the dimension of data (matrices)

Now we look at another dimensionality reduction technique: principal component analysis (PCA)

PCA is often used as a blackbox technique, here we take a look at the math behind it

What is PCA?

Linear algebraic technique

Helps reduce a complex dataset to a lower dimensional one

Non-parametric method: does not assume anything about data distribution (distribution from the statistical point of view)

A 2D example

Dataset (link) of height and weight of some hockey players

```
# From https://figshare.com/ndownloader/files/5303173
data = read.csv("https://github.com/julien-arino/math-of-data-science/raw/re
dim(data)

## [1] 6292   13
```

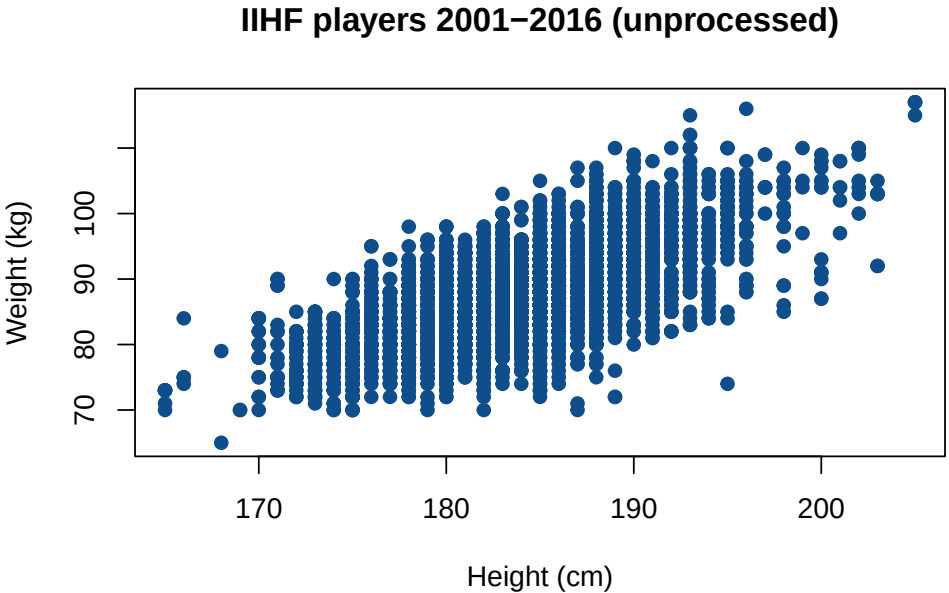
In case you are wondering, this is a database of ice hockey players at IIHF world championships, 2001-2016, assembled by the dataset's author

See some comments here

```r
head(data, n=3)
```

```
##   year country no                name position side height weight      bi
## 1 2001     RUS 10    tverdovsky oleg        D    L    185     84 1976-05
## 2 2001     RUS  2  vichnevsky vitali        D    L    188     86 1980-03
## 3 2001     RUS 26 petrochinin evgeni        D    L    182     95 1976-02
##                      club      age cohort      bmi
## 1   anaheim mighty ducks  24.95277   1976 24.54346
## 2   anaheim mighty ducks  21.11978   1980 24.33228
## 3 severstal cherepovetal  25.22930   1976 28.68011
```

As usual, it is a good idea to plot this to get a sense of the lay of the land


IIHF players 2001−2016 (unprocessed)
Weight (kg)
70
80
90
100
170
180
190
200
Height (cm)

The author of the study is interested in the evolution of weights, so it is likely that the same person will be in the dataset several times

Let us check this: first check below will be `FALSE` if the number of unique names does not match the number of rows in the dataset

```
length(unique(data$name)) == dim(data)[1]

## [1] FALSE

length(unique(data$name))

## [1] 3278

dim(data)[1]

## [1] 6292
```

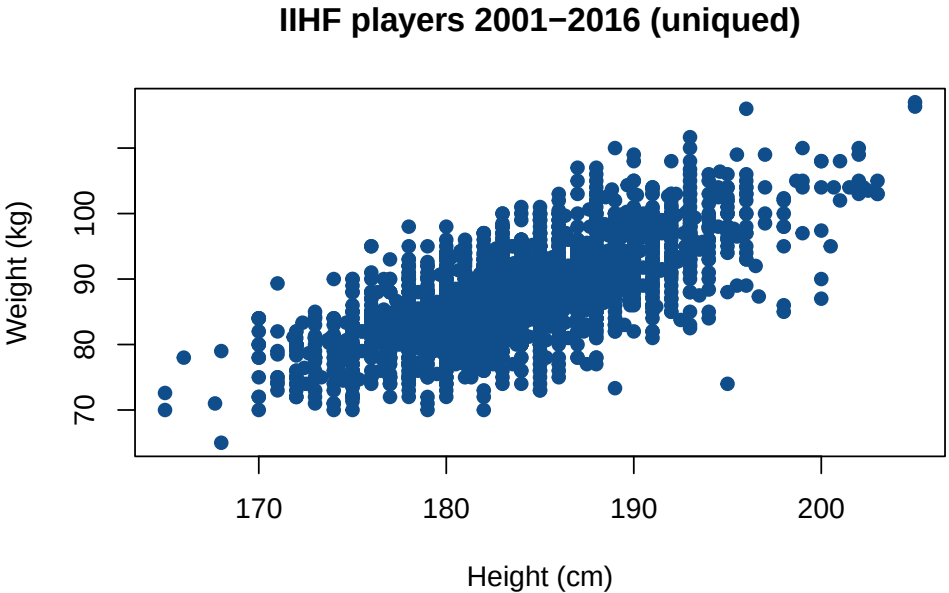
We are not interested in the evolution of weights, so we simplify: if more than one record for someone, take average of recorded weights and heights

To be extra careful, could check as well that there are no major variations on player height (homonymies?)

```
data_simplified = data.frame(name = unique(data$name))
w = c()
h = c()
for (n in data_simplified$name) {
    tmp = data[which(data$name == n),]
    h = c(h, mean(tmp$height))
    w = c(w, mean(tmp$weight))
}
data_simplified$weight = w
data_simplified$height = h
```

```r
data = data_simplified
head(data_simplified, n = 6)
```

```
##                   name weight height
## 1    tverdovsky oleg    84.0  185.0
## 2  vichnevsky vitali    86.0  188.0
## 3 petrochinin evgeni    95.0  182.0
## 4    zhdan alexander    85.5  178.5
## 5    orekhovsky oleg    88.0  175.0
## 6      zhukov sergei    92.5  193.0
```


IIHF players 2001−2016 (uniqued)
Weight (kg)
70
80
90
100
170
180
190
200
Height (cm)

Centre the data

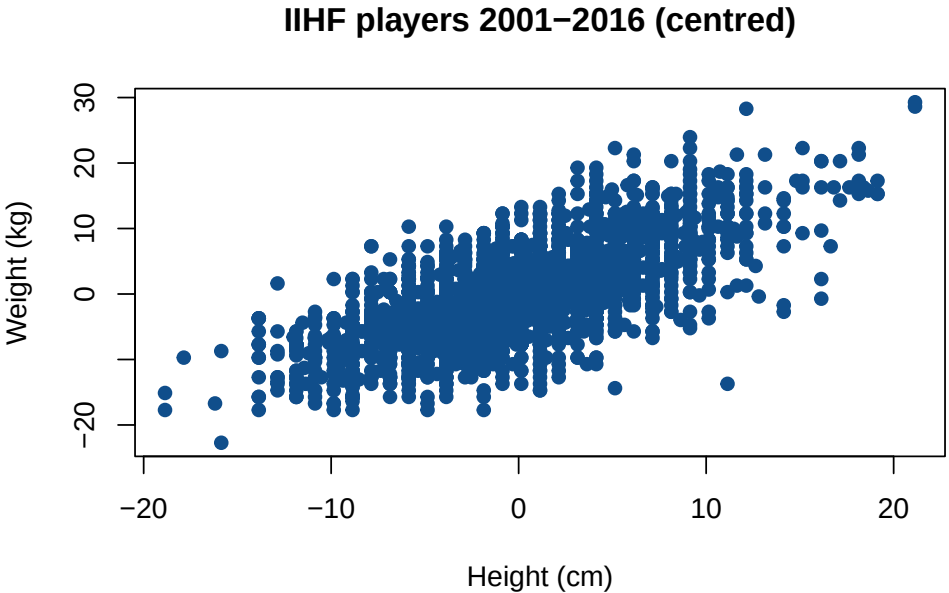
```
mean(data$weight)

## [1] 87.71555

mean(data$height)

## [1] 183.8596

data$weight.c = data$weight-mean(data$weight)
data$height.c = data$height-mean(data$height)
```


IIHF players 2001−2016 (centred)
Weight (kg)
30
20
10
0
−20
−20
−10
0
10
20
Height (cm)

Setting things up

Each player is a row in the matrix (an *observation*), each variable (*height* and *weight*) is a column

After deduplication, we have an 3278×2 matrix (actually, 3278×4 if we consider the uncentred and centred variables, but we will use one or the other, not both uncentred and centred)

We want to find what carries the most information

For this, we are going to project the information in a new basis in which the first "dimension" will carry most information (in a sense we'll define later), the second dimension will carry a little less, etc.

In order to do so, we need to learn how to change bases

In the following,

$$[\boldsymbol{x}]_{\mathcal{B}}$$

denotes the coordinates of $\boldsymbol{x}$ in the basis $\mathcal{B} = \{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n\}$, i.e., $[\boldsymbol{x}]_{\mathcal{B}} = (b_1, \ldots, b_n)^T$ such that $\boldsymbol{x} = b_1\boldsymbol{u}_1 + \cdots + b_n\boldsymbol{u}_n$

The aim of a change of basis is to express vectors in another coordinate system (another basis) $\mathcal{C} = \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$

We do so by finding a matrix allowing to move from one basis to another

Change of basis

Definition 80 (Change of basis matrix)

Let $\mathcal{B} = \{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n\}$ and $\mathcal{C} = \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$ be bases of vector space V

The **change of basis matrix** $P_{\mathcal{C} \leftarrow \mathcal{B}} \in \mathcal{M}_n$ from basis $\mathcal{B}$ to $\mathcal{C}$ is

$$P_{\mathcal{C} \leftarrow \mathcal{B}} = [[\boldsymbol{u}_1]_{\mathcal{C}} \cdots [\boldsymbol{u}_n]_{\mathcal{C}}]$$

i.e., its columns are the coordinate vectors $[\boldsymbol{u}_1]_{\mathcal{C}}, \ldots, [\boldsymbol{u}_n]_{\mathcal{C}}$ of vectors in $\mathcal{B}$ with respect to $\mathcal{C}$

Example – $\mathcal{B}$ is the canonical basis of $\mathbb{R}^3$

Let $\mathcal{B} = \{\boldsymbol{i}, \boldsymbol{j}, \boldsymbol{k}\}$ be the canonical basis of $\mathbb{R}^3$ and $\mathcal{C} = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$, with

$$\boldsymbol{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \boldsymbol{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \boldsymbol{v}_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

To find $P_{\mathcal{C}\leftarrow\mathcal{B}} = [[\boldsymbol{i}]_{\mathcal{C}} \;\; [\boldsymbol{j}]_{\mathcal{C}} \;\; [\boldsymbol{k}]_{\mathcal{C}}]$, express each canonical vector in terms of $\mathcal{C}$:

$$\boldsymbol{i} = \boldsymbol{v}_1, \quad \boldsymbol{j} = \boldsymbol{v}_2 - \boldsymbol{v}_1, \quad \boldsymbol{k} = \boldsymbol{v}_3 - \boldsymbol{v}_2$$

$$\implies [\boldsymbol{i}]_{\mathcal{C}} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad [\boldsymbol{j}]_{\mathcal{C}} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \quad [\boldsymbol{k}]_{\mathcal{C}} = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$$

The change of basis matrix from $\mathcal{B}$ to $\mathcal{C}$ is

$$P_{\mathcal{C}\leftarrow\mathcal{B}} = [[\boldsymbol{i}]_{\mathcal{C}} \;\; [\boldsymbol{j}]_{\mathcal{C}} \;\; [\boldsymbol{k}]_{\mathcal{C}}] = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

Example – $\mathcal{C}$ is the canonical basis of $\mathbb{R}^3$

Now consider the reverse: let $\mathcal{B} = \{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\}$ with

$$\boldsymbol{u}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \boldsymbol{u}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \boldsymbol{u}_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

and let $\mathcal{C} = \{\boldsymbol{i}, \boldsymbol{j}, \boldsymbol{k}\}$ be the canonical basis of $\mathbb{R}^3$

Since $\mathcal{C}$ is the canonical basis, the coordinates in $\mathcal{C}$ are simply the vector itself:

$$[\boldsymbol{u}_1]_{\mathcal{C}} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad [\boldsymbol{u}_2]_{\mathcal{C}} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad [\boldsymbol{u}_3]_{\mathcal{C}} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Thus, the change of basis matrix from $\mathcal{B}$ to the canonical basis $\mathcal{C}$ is

$$P_{\mathcal{C} \leftarrow \mathcal{B}} = [[\boldsymbol{u}_1]_{\mathcal{C}} \ \ [\boldsymbol{u}_2]_{\mathcal{C}} \ \ [\boldsymbol{u}_3]_{\mathcal{C}}] = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

Theorem 81

$\mathcal{B} = \{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n\}$ and $\mathcal{C} = \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$ bases of vector space V and $P_{\mathcal{C}\leftarrow\mathcal{B}}$ a change of basis matrix from $\mathcal{B}$ to $\mathcal{C}$

1. $\forall \boldsymbol{x} \in V$, $P_{\mathcal{C}\leftarrow\mathcal{B}}[\boldsymbol{x}]_{\mathcal{B}} = [\boldsymbol{x}]_{\mathcal{C}}$
2. $P_{\mathcal{C}\leftarrow\mathcal{B}}$ *s.t.* $\forall \boldsymbol{x} \in V$, $P_{\mathcal{C}\leftarrow\mathcal{B}}[\boldsymbol{x}]_{\mathcal{B}} = [\boldsymbol{x}]_{\mathcal{C}}$ *is* ***unique***
3. $P_{\mathcal{C}\leftarrow\mathcal{B}}$ *invertible and* $P^{-1}_{\mathcal{C}\leftarrow\mathcal{B}} = P_{\mathcal{B}\leftarrow\mathcal{C}}$

Example – Linking the two directions

Let $\mathcal{B} = \{\boldsymbol{i}, \boldsymbol{j}, \boldsymbol{k}\}$ be the canonical basis and $\mathcal{C} = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$ with

$$\boldsymbol{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \boldsymbol{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \boldsymbol{v}_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

From our two previous examples:

$$P_{\mathcal{C}\leftarrow\mathcal{B}} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{and} \quad P_{\mathcal{B}\leftarrow\mathcal{C}} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

By item 3 in Theorem **??**, $P^{-1}_{\mathcal{C}\leftarrow\mathcal{B}} = P_{\mathcal{B}\leftarrow\mathcal{C}}$. Indeed:

$$P_{\mathcal{C}\leftarrow\mathcal{B}} P_{\mathcal{B}\leftarrow\mathcal{C}} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \mathbb{I}$$

Since $P_{\mathcal{B}\leftarrow\mathcal{C}} = [\boldsymbol{v}_1 \;\; \boldsymbol{v}_2 \;\; \boldsymbol{v}_3]$ is immediate, we have $P_{\mathcal{C}\leftarrow\mathcal{B}} = [\boldsymbol{v}_1 \;\; \boldsymbol{v}_2 \;\; \boldsymbol{v}_3]^{-1}$

Row-reduction method for changing bases

Theorem 82

$\mathcal{B} = \{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n\}$ and $\mathcal{C} = \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$ bases of vector space V. Let $\mathcal{E}$ be any basis for V,

$$B = [[\boldsymbol{u}_1]_{\mathcal{E}}, \ldots, [\boldsymbol{u}_n]_{\mathcal{E}}] \ \textit{and}\ C = [[\boldsymbol{v}_1]_{\mathcal{E}}, \ldots, [\boldsymbol{v}_n]_{\mathcal{E}}]$$

and let $[C|B]$ be the augmented matrix constructed using C and B. Then

$$RREF\,([C|B]) = [\mathbb{I}|P_{\mathcal{C}\leftarrow\mathcal{B}}]$$

If working in $\mathbb{R}^n$, this is quite useful with $\mathcal{E}$ the standard basis of $\mathbb{R}^n$ (it does not matter if $\mathcal{B} = \mathcal{E}$)

So the question now becomes

How do we find what new basis to look at our data in?

(Changing the basis does not change the data, just the view you have of it)

(Think of what happens when you do a headstand.. your up becomes down, your right and left switch, but the world does not change, just your view of it)

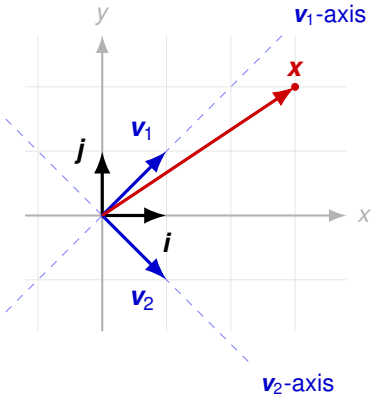
(Changes of bases are *fundamental* operations in Science)

Example: change of basis

Problem: Find the change of basis matrix from basis $\mathcal{B}$ to basis $\mathcal{C}$ in $\mathbb{R}^2$, where

$$\mathcal{B} = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\} \quad \text{and} \quad \mathcal{C} = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}$$

Then use this matrix to find the coordinates of $\boldsymbol{x} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$ in basis $\mathcal{C}$



Step 1 – Set up the matrices

- $\mathcal{B} = \{\boldsymbol{i}, \boldsymbol{j}\}$ is the standard basis of $\mathbb{R}^2$, so using $\mathcal{E} = \mathcal{B}$, we have

$$[\boldsymbol{i}]_{\mathcal{E}} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad [\boldsymbol{j}]_{\mathcal{E}} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

- For $\mathcal{C}$:

$$[\boldsymbol{v}_1]_{\mathcal{E}} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad [\boldsymbol{v}_2]_{\mathcal{E}} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Step 2 – Row reduce

Using Theorem **??**, we form the augmented matrix $[C|B]$:

$$[C|B] = \left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 1 & -1 & 0 & 1 \end{array}\right]$$

Row reduce to RREF:

$$\left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 1 & -1 & 0 & 1 \end{array}\right] \xrightarrow{R_2 \leftarrow R_2 - R_1} \left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 0 & -2 & -1 & 1 \end{array}\right]$$

$$\xrightarrow{R_2 \leftarrow -\frac{1}{2}R_2} \left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 0 & 1 & \frac{1}{2} & -\frac{1}{2} \end{array}\right]$$

$$\xrightarrow{R_1 \leftarrow R_1 - R_2} \left[\begin{array}{cc|cc} 1 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & \frac{1}{2} & -\frac{1}{2} \end{array}\right]$$

Step 3 – Extract the change of basis matrix

From the RREF form $[\mathbb{I}|P_{\mathcal{C}\leftarrow\mathcal{B}}]$, we get:

$$P_{\mathcal{C}\leftarrow\mathcal{B}} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}$$

Verification: Let's check that this matrix works correctly.

- $[\boldsymbol{i}]_{\mathcal{C}} = P_{\mathcal{C}\leftarrow\mathcal{B}}[\boldsymbol{i}]_{\mathcal{B}} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$
- Check: $\frac{1}{2}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{1}{2}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

Step 4 – Coordinates of $\boldsymbol{x}$ in basis $\mathcal{C}$

Now we find $[\boldsymbol{x}]_{\mathcal{C}}$ for $\boldsymbol{x} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$:

$$[\boldsymbol{x}]_{\mathcal{C}} = P_{\mathcal{C} \leftarrow \mathcal{B}} [\boldsymbol{x}]_{\mathcal{B}} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} \frac{5}{2} \\ \frac{1}{2} \end{pmatrix}$$

Verification: Check that this gives us back the original vector:

$$\frac{5}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} \frac{5}{2} + \frac{1}{2} \\ \frac{5}{2} - \frac{1}{2} \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

Answer: The coordinates of $\boldsymbol{x} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$ in basis $\mathcal{C}$ are $[\boldsymbol{x}]_{\mathcal{C}} = \begin{pmatrix} \frac{5}{2} \\ \frac{1}{2} \end{pmatrix}$

Alternative method – Direct calculation

We could also solve this directly by setting up the system:

$$\boldsymbol{x} = c_1 \boldsymbol{v}_1 + c_2 \boldsymbol{v}_2$$

$$\begin{pmatrix} 3 \\ 2 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

This gives us the system:

$$\begin{aligned} c_1 + c_2 &= 3 \\ c_1 - c_2 &= 2 \end{aligned}$$

Solving: $c_1 = \frac{5}{2}$, $c_2 = \frac{1}{2}$

This confirms our result: $[\boldsymbol{x}]_{\mathcal{C}} = \begin{pmatrix} \frac{5}{2} \\ \frac{1}{2} \end{pmatrix}$

Computation in R – Finding $P_{\mathcal{C}\leftarrow\mathcal{B}}$

Set up $C = [\boldsymbol{v}_1, \boldsymbol{v}_2]$, $B = [\boldsymbol{i}, \boldsymbol{j}]$, form the augmented matrix $[C|B]$ and row reduce:

```
C = matrix(c(1, 1, 1, -1), nrow = 2)
B = diag(2)
CB = cbind(C, B)
(R = pracma::rref(CB))
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    1    0  0.5  0.5
## [2,]    0    1  0.5 -0.5
```

```
(P = R[, 3:4])
```

```
##      [,1] [,2]
## [1,]  0.5  0.5
## [2,]  0.5 -0.5
```

Computation in R – Coordinates of $\boldsymbol{x}$

Now compute $[\boldsymbol{x}]_{\mathcal{C}} = P_{\mathcal{C}\leftarrow\mathcal{B}}[\boldsymbol{x}]_{\mathcal{B}}$ and verify that $C[\boldsymbol{x}]_{\mathcal{C}} = \boldsymbol{x}$:

```
x = c(3, 2)
(x_C = P %*% x)
```

```
##      [,1]
## [1,]  2.5
## [2,]  0.5
```

```
# Verification: reconstruct x in the canonical basis
C %*% x_C
```

```
##      [,1]
## [1,]    3
## [2,]    2
```

Example – A trajectory in $\mathbb{R}^2$

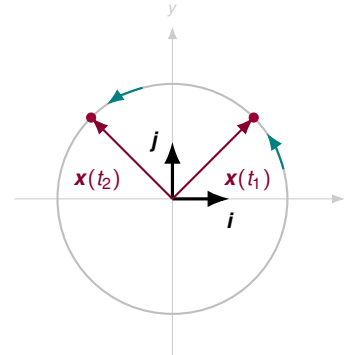
Let $\mathcal{B} = \{\boldsymbol{i}, \boldsymbol{j}\}$ be the canonical basis of $\mathbb{R}^2$.

Consider a trajectory along the unit circle:

$$\boldsymbol{x}(t) = \begin{pmatrix} \cos t \\ \sin t \end{pmatrix}, \quad t \in \mathbb{R}$$

This is a function $\boldsymbol{x} : \mathbb{R} \to \mathbb{R}^2$ that we call a **parametric equation**:

- t is the parameter (often interpreted as time)
- For each t, $\boldsymbol{x}(t)$ is a point in $\mathbb{R}^2$; as t varies, $\boldsymbol{x}(t)$ traces out the trajectory



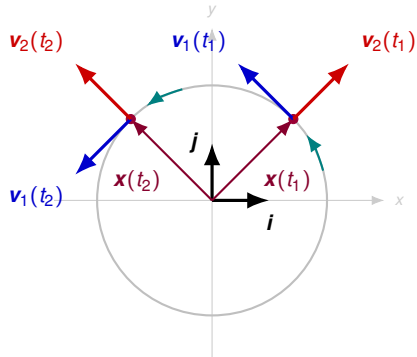
A moving basis along the trajectory

At each point $\boldsymbol{x}(t)$ on the trajectory, define a moving basis $\mathcal{C}(t) = \{\boldsymbol{v}_1(t), \boldsymbol{v}_2(t)\}$:

- $\boldsymbol{v}_1(t) = \boldsymbol{x}'(t) = \begin{pmatrix} -\sin t \\ \cos t \end{pmatrix}$ (unit tangent)
- $\boldsymbol{v}_2(t) = \begin{pmatrix} \cos t \\ \sin t \end{pmatrix}$ (outward unit normal)

Since $\boldsymbol{v}_1(t) \bullet \boldsymbol{v}_2(t) = 0$ and $\|\boldsymbol{v}_1\| = \|\boldsymbol{v}_2\| = 1$, $\mathcal{C}(t)$ is an orthonormal basis for all t

As t changes, the basis $\{\boldsymbol{v}_1(t), \boldsymbol{v}_2(t)\}$ moves and rotates along with $\boldsymbol{x}(t)$



Coordinates in the moving basis

The change of basis matrix from $\mathcal{C}(t)$ to the canonical basis $\mathcal{B}$ is

$$P_{\mathcal{B}\leftarrow\mathcal{C}(t)} = [\boldsymbol{v}_1(t) \;\; \boldsymbol{v}_2(t)] = \begin{pmatrix} -\sin t & \cos t \\ \cos t & \sin t \end{pmatrix}$$

Since $P_{\mathcal{B}\leftarrow\mathcal{C}(t)}$ is symmetric and orthogonal, the change of basis matrix to $\mathcal{C}(t)$ is:

$$P_{\mathcal{C}(t)\leftarrow\mathcal{B}} = P^{-1}_{\mathcal{B}\leftarrow\mathcal{C}(t)} = P^{T}_{\mathcal{B}\leftarrow\mathcal{C}(t)} = \begin{pmatrix} -\sin t & \cos t \\ \cos t & \sin t \end{pmatrix}$$

In the moving basis $\mathcal{C}(t)$, the coordinates of position and velocity are:

$$[\boldsymbol{x}(t)]_{\mathcal{C}(t)} = P_{\mathcal{C}(t)\leftarrow\mathcal{B}} \begin{pmatrix} \cos t \\ \sin t \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \qquad [\boldsymbol{x}'(t)]_{\mathcal{C}(t)} = P_{\mathcal{C}(t)\leftarrow\mathcal{B}} \begin{pmatrix} -\sin t \\ \cos t \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

In the local frame, the particle is always at distance 1 along the normal axis with velocity 1 along the tangent

Choosing the inward normal $-\boldsymbol{v}_2(t)$ *simply gives* $[\boldsymbol{x}(t)]_{\mathcal{C}(t)} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$

Why probability?

We said earlier that we would look for a basis in which the first dimension carries most information

But how do we define *information*?

We use concepts from probability and statistics to do so

A good measure of information is *variance* (how much data varies around the mean)

Brief “review” of some probability concepts

Proper definition of *probability* requires to use *measure theory*.. will not get into details here

A **random variable** X is a *measurable* function $X : \Omega \to E$, where Ω is a set of outcomes (*sample space*) and E is a measurable space

$\mathbb{P}(X \in S \subseteq E) = \mathbb{P}(\omega \in \Omega | X(\omega) \in S)$

Distribution function of a r.v., $F(x) = \mathbb{P}(X \leq x)$, describes the distribution of a r.v.

R.v. can be discrete or continuous or .. other things.

Variance

Definition 83 (Variance)

Let X be a random variable. The **variance** of X is given by

$$\operatorname{Var} X = E\left[(X - E(X))^2\right]$$

where E is the expected value

Covariance

Definition 84 (Covariance)

Let X, Y be jointly distributed random variables. The **covariance** of X and Y is given by

$$\operatorname{cov}(X, Y) = E\left[(X - E(X))\,(Y - E(Y))\right]$$

Note that $\operatorname{cov}(X, X) = E\left[(X - E(X))^2\right] = \operatorname{Var} X$

In practice: “true law” versus “observation”

In statistics: we reason on the *true law* of distributions, but we usually have only access to a sample

We then use **estimators** to .. estimate the value of a parameter, e.g., the mean, variance and covariance

Definition 85 (Unbiased estimators of the mean and variance)

Let $x_1, \ldots, x_n$ be data points (the *sample*) and

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

be the **mean** of the data. An unbiased estimator of the variance of the sample is

$$\sigma^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$$

Unbiased: the expected value of the estimator equals the true parameter value ($E(\bar{x}) = E(X)$ and $E(\sigma^2) = \text{Var } X$): on average, it does not systematically overestimate or underestimate the parameter

Variance of hockey players' height and weight

We compute the sample variance in R using the function `var`:

```
var(data$height)

## [1] 29.66176

var(data$weight)

## [1] 47.81112
```

The variance is about 29.7 cm^2 for height and 47.8 kg^2 for weight

Normalising does not change the variance

Centring (normalising) shifts data by a constant, leaving spread unchanged:

$$\text{Var}\,(X - c) = \text{Var}\,X$$

Indeed, computing the variance using the centred values:

```
var(data$height.c)
```

```
## [1] 29.66176
```

```
var(data$weight.c)
```

```
## [1] 47.81112
```

The variance is identical to that of the uncentred data

Definition 86 (Unbiased estimator of the covariance - 2D case)

Let $(x_1, y_1), \ldots, (x_n, y_n)$ be data points,

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \text{ and } \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$$

be the means of the data. An unbiased estimator of the covariance of the sample is

$$\text{cov}(x, y) = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

Computation of covariance in R – “By hand” and with a function

We compute the means and sum directly from the definition:

```
n = length(data$height)
x_bar = sum(data$height) / n
y_bar = sum(data$weight) / n
sum((data$height - x_bar) * (data$weight - y_bar)) / (n - 1)

## [1] 26.63506

cov(data$height, data$weight)

## [1] 26.63506
```

The covariance is about 26.6 cm · kg, matching R’s built-in `cov`

What does covariance do?

Variance explains how data disperses around the mean, in a 1-D case

Covariance measures the relationship between two dimensions. E.g., height and weight

More than the exact value, the sign is important:

- $\text{cov}(X, Y) > 0$: both dimensions change in the same "direction"; e.g., larger height usually means higher weight
- $\text{cov}(X, Y) < 0$: both dimensions change in reverse directions; e.g., time spent on social media and performance in this class
- $\text{cov}(X, Y) = 0$: the dimensions are independent from one another; e.g., sex/gender and "intelligence"

The covariance matrix

Definition 87 (Covariance matrix)

Suppose p random variables $X_1, \ldots, X_p$. Then the covariance matrix is the symmetric matrix

$$\begin{pmatrix} \operatorname{Var} X_1 & \operatorname{cov}(X_1, X_2) & \cdots & \operatorname{cov}(X_1, X_p) \\ \operatorname{cov}(X_1, X_2) & \operatorname{Var} X_2 & \cdots & \operatorname{cov}(X_2, X_p) \\ \vdots & \vdots & & \vdots \\ \operatorname{cov}(X_1, X_p) & \operatorname{cov}(X_2, X_p) & \cdots & \operatorname{Var} X_p \end{pmatrix}$$

The actual form of the matrix is below but the form above stems from the properties of covariance

$$\begin{pmatrix} \operatorname{cov}(X_1, X_1) & \operatorname{cov}(X_1, X_2) & \cdots & \operatorname{cov}(X_1, X_p) \\ \operatorname{cov}(X_2, X_1) & \operatorname{cov}(X_2, X_2) & \cdots & \operatorname{cov}(X_2, X_p) \\ \vdots & \vdots & & \vdots \\ \operatorname{cov}(X_p, X_1) & \operatorname{cov}(X_p, X_2) & \cdots & \operatorname{cov}(X_p, X_p) \end{pmatrix}$$

Sample covariance matrix

$X \in \mathcal{M}_{np}$ the data (n observations, p variables), then the (sample) covariance matrix S_X takes the form

$$S_X = \frac{1}{n-1} X^T X$$

where the data is centred!

Sometimes you will see $S_X = 1/(n-1) XX^T$. This is for matrices with observations in columns and variables in rows. Just remember that you want the covariance matrix to have size the number of variables, not observations, this will give you the order in which to take the product

Why is $S_X = \frac{1}{n-1} X^T X$? – Data matrix setup

Let us prove why the sample covariance matrix takes this matrix form

Suppose we have n observations of p variables $X_1, \ldots, X_p$

Organise the data into a matrix $X \in \mathcal{M}_{np}$ where each row is an observation and each column is a variable:

$$X = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix} = \begin{pmatrix} \boldsymbol{c}_1 & \boldsymbol{c}_2 & \cdots & \boldsymbol{c}_p \end{pmatrix}$$

where column $\boldsymbol{c}_j = (x_{1j}, x_{2j}, \ldots, x_{nj})^T \in \mathbb{R}^n$ contains all n measurements of variable X_j

Assume the data is **centred**: $\bar{x}_j = \frac{1}{n} \sum_{i=1}^{n} x_{ij} = 0$ for all $j = 1, \ldots, p$

Why is $S_X = \frac{1}{n-1} X^T X$? – Sample covariances

Recall the definition of the sample covariance between variables X_j and X_k:

$$\operatorname{cov}(X_j, X_k) = \frac{1}{n-1} \sum_{i=1}^{n} (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k)$$

Since the data is centred ($\bar{x}_j = \bar{x}_k = 0$), this simplifies to:

$$\operatorname{cov}(X_j, X_k) = \frac{1}{n-1} \sum_{i=1}^{n} x_{ij} x_{ik}$$

In particular, when $j = k$, we obtain the sample variance of variable X_j:

$$\operatorname{Var} X_j = \operatorname{cov}(X_j, X_j) = \frac{1}{n-1} \sum_{i=1}^{n} x_{ij}^2$$

Notice: $\sum_{i=1}^{n} x_{ij} x_{ik} = \boldsymbol{c}_j^T \boldsymbol{c}_k$, the dot product of column $\boldsymbol{c}_j$ and column $\boldsymbol{c}_k$!

Why is $S_X = \frac{1}{n-1} X^T X$? – Matrix product $X^T X$

Now consider the matrix product $X^T X$:

$X^T \in \mathcal{M}_{pn}$ has the columns of X as its rows:

$$X^T = \begin{pmatrix} \boldsymbol{c}_1^T \\ \boldsymbol{c}_2^T \\ \vdots \\ \boldsymbol{c}_p^T \end{pmatrix} = \begin{pmatrix} x_{11} & x_{21} & \cdots & x_{n1} \\ x_{12} & x_{22} & \cdots & x_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ x_{1p} & x_{2p} & \cdots & x_{np} \end{pmatrix}$$

Since $X^T \in \mathcal{M}_{pn}$ and $X \in \mathcal{M}_{np}$, the product $X^T X$ is a $p \times p$ matrix

Its (j, k)-entry is the product of row j of X^T and column k of X:

$$(X^T X)_{jk} = \boldsymbol{c}_j^T \boldsymbol{c}_k = \sum_{i=1}^{n} x_{ij} x_{ik}$$

Why is $S_X = \frac{1}{n-1} X^T X$? – Conclusion

Dividing by $n-1$, each entry of $\frac{1}{n-1} X^T X$ is:

$$\left(\frac{1}{n-1} X^T X \right)_{jk} = \frac{1}{n-1} \sum_{i=1}^{n} x_{ij} x_{ik} = \begin{cases} \text{Var } X_j & \text{if } j = k \\ \text{cov}(X_j, X_k) & \text{if } j \neq k \end{cases}$$

Thus, writing out the matrix:

$$\frac{1}{n-1} X^T X = \begin{pmatrix} \text{Var } X_1 & \text{cov}(X_1, X_2) & \cdots & \text{cov}(X_1, X_p) \\ \text{cov}(X_2, X_1) & \text{Var } X_2 & \cdots & \text{cov}(X_2, X_p) \\ \vdots & \vdots & \ddots & \vdots \\ \text{cov}(X_p, X_1) & \text{cov}(X_p, X_2) & \cdots & \text{Var } X_p \end{pmatrix} = S_X$$

This proves that $S_X = \frac{1}{n-1} X^T X$ for centred data

What if observations are in columns?

Suppose data is stored with observations in columns and variables in rows:

$$Y = X^T \in \mathcal{M}_{pn}$$

where row j of Y is $\boldsymbol{c}_j^T = (x_{1j}, \ldots, x_{nj}) \in \mathbb{R}^{1\times n}$ (variable X_j)

Then $YY^T \in \mathcal{M}_p$, and its (j,k)-entry is:

$$(YY^T)_{jk} = \boldsymbol{c}_j^T \boldsymbol{c}_k = \sum_{i=1}^{n} x_{ij}x_{ik}$$

Dividing by $n-1$:

$$S_Y = \frac{1}{n-1} YY^T = S_X$$

The resulting sample covariance matrix is identical

What if the data is not centred beforehand?

Suppose $\tilde{X} \in \mathcal{M}_{np}$ is **uncentred**

Let $\mathbf{1} = (1, 1, \ldots, 1)^T \in \mathbb{R}^n$. The vector of column means is $\bar{\boldsymbol{x}}^T = \frac{1}{n}\mathbf{1}^T\tilde{X} \in \mathbb{R}^{1\times p}$

Subtracting the means from each row of $\tilde{X}$:

$$X = \tilde{X} - \mathbf{1}\bar{\boldsymbol{x}}^T = \tilde{X} - \frac{1}{n}\mathbf{1}\mathbf{1}^T\tilde{X} = \left(I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^T\right)\tilde{X} = C_n\tilde{X}$$

where $C_n = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^T \in \mathcal{M}_n$ is the **centering matrix**

C_n is symmetric ($C_n^T = C_n$) and idempotent ($C_n^2 = C_n$), so:

$$S_X = \frac{1}{n-1}X^TX = \frac{1}{n-1}\tilde{X}^TC_n^TC_n\tilde{X} = \frac{1}{n-1}\tilde{X}^TC_n\tilde{X}$$

Covariance matrix of hockey players in R

When passed a data frame or matrix, `cov` computes the covariance matrix:

```
cov(data[, c("height", "weight")])

##           height   weight
## height 29.66176 26.63506
## weight 26.63506 47.81112
```

The diagonal entries are the variances (Var (height) ≈ 29.7, Var (weight) ≈ 47.8) and the off-diagonal entries are cov(height, weight) ≈ 26.6

Setting things up

I will use notation (mostly) as in Joliffe's *Principal Component Analysis* (PDF of older version available for free from UofM Libraries)

$\boldsymbol{x} = (x_1, \ldots, x_p)$ vector of p random variables

We seek a linear function $\boldsymbol{\alpha}_1^T \boldsymbol{x}$ with maximum variance, where $\boldsymbol{\alpha}_1 = (\alpha_{11}, \ldots, \alpha_{1p})$, i.e.,

$$\boldsymbol{\alpha}_1^T \boldsymbol{x} = \sum_{j=1}^{p} \alpha_{1j} x_j$$

Then we seek a linear function $\boldsymbol{\alpha}_2^T \boldsymbol{x}$ with maximum variance, uncorrelated to $\boldsymbol{\alpha}_1^T \boldsymbol{x}$

And we continue...

At *k*th stage, we find a linear function $\boldsymbol{\alpha}_k^T \boldsymbol{x}$ with maximum variance, uncorrelated to $\boldsymbol{\alpha}_1^T \boldsymbol{x}, \ldots, \boldsymbol{\alpha}_{k-1}^T \boldsymbol{x}$

$\boldsymbol{\alpha}_i^T \boldsymbol{x}$ is the *i*th **principal component** (PC)

Case of known covariance matrix

Suppose we know Σ, covariance matrix of $\boldsymbol{x}$ (i.e., typically: we know $\boldsymbol{x}$)

Then the kth PC is

$$z_k = \boldsymbol{\alpha}_k^T \boldsymbol{x}$$

where $\boldsymbol{\alpha}_k$ is an eigenvector of Σ corresponding to the kth largest eigenvalue λ_k

If, additionally, $\|\boldsymbol{\alpha}_k\| = \boldsymbol{\alpha}_k^T \boldsymbol{\alpha} = 1$, then $\lambda_k = \text{Var } z_k$

Why is that?

Let us start with

$$\boldsymbol{\alpha}_1^T \boldsymbol{x}$$

We want maximum variance, where $\boldsymbol{\alpha}_1 = (\alpha_{11}, \ldots, \alpha_{1p})$, i.e.,

$$\boldsymbol{\alpha}_1^T \boldsymbol{x} = \sum_{j=1}^{p} \alpha_{1j} x_j$$

with the constraint that $\|\boldsymbol{\alpha}_1\| = 1$

We have

$$\text{Var } \boldsymbol{\alpha}_1^T \boldsymbol{x} = \boldsymbol{\alpha}_1^T \Sigma \boldsymbol{\alpha}_1$$

Objective

We want to maximise Var $\boldsymbol{\alpha}_1^T \boldsymbol{x}$, i.e.,

$$\boldsymbol{\alpha}_1^T \Sigma \boldsymbol{\alpha}_1$$

under the constraint that $\|\boldsymbol{\alpha}_1\| = 1$

$\Longrightarrow$ use **Lagrange multipliers**

Maximisation using Lagrange multipliers

(A.k.a. super-brief intro to multivariable calculus)

We want the max of $f(x_1, \ldots, x_n)$ under the constraint $g(x_1, \ldots, x_n) = k$

1. Solve

$$\nabla f(x_1, \ldots, x_n) = \lambda \nabla g(x_1, \ldots, x_n)$$
$$g(x_1, \ldots, x_n) = k$$

   where $\nabla = (\frac{\partial}{\partial x_1}, \ldots, \frac{\partial}{\partial x_n})$ is the **gradient operator**
2. Plug all solutions into $f(x_1, \ldots, x_n)$ and find maximum values (provided values exist and $\nabla g \neq \mathbf{0}$ there)

λ is the **Lagrange multiplier**

The gradient

(Continuing our super-brief intro to multivariable calculus)

$f : \mathbb{R}^n \to \mathbb{R}$ function of several variables, $\nabla = \left(\frac{\partial}{\partial x_1}, \ldots, \frac{\partial}{\partial x_n}\right)$ the gradient operator

Then

$$\nabla f = \left(\frac{\partial}{\partial x_1} f, \ldots, \frac{\partial}{\partial x_n} f\right)$$

So ∇f is a *vector-valued* function, $\nabla f : \mathbb{R}^n \to \mathbb{R}^n$; also written as

$$\nabla f = f_{x_1}(x_1, \ldots, x_n)\boldsymbol{e}_1 + \cdots f_{x_n}(x_1, \ldots, x_n)\boldsymbol{e}_n$$

where f_{x_i} is the partial derivative of f with respect to x_i and $\{\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n\}$ is the standard basis of $\mathbb{R}^n$

Bear with me..

(You may experience a brief period of discomfort)

$\boldsymbol{\alpha}_1^T \Sigma \boldsymbol{\alpha}_1$ and $\|\boldsymbol{\alpha}_1\|^2 = \boldsymbol{\alpha}_1^T \boldsymbol{\alpha_1}$ are functions of $\boldsymbol{\alpha}_1 = (\alpha_{11}, \ldots, \alpha_{1p})$

In the notation of the previous slide, we want the max of

$$f(\alpha_{11}, \ldots, \alpha_{1p}) := \boldsymbol{\alpha}_1^T \Sigma \boldsymbol{\alpha}_1$$

under the constraint that

$$g(\alpha_{11}, \ldots, \alpha_{1p}) := \boldsymbol{\alpha}_1^T \boldsymbol{\alpha_1} = 1$$

and with gradient operator

$$\nabla = \left(\frac{\partial}{\partial \alpha_{11}}, \ldots, \frac{\partial}{\partial \alpha_{1p}} \right)$$

Effect of ∇ on g

g is easiest to see:

$$\begin{aligned}
\nabla g(\alpha_{11}, \ldots, \alpha_{1p}) &= \left(\frac{\partial}{\partial \alpha_{11}}, \ldots, \frac{\partial}{\partial \alpha_{1p}} \right) (\alpha_{11}, \ldots, \alpha_{1p}) \begin{pmatrix} \alpha_{11} \\ \vdots \\ \alpha_{1p} \end{pmatrix} \\
&= \left(\frac{\partial}{\partial \alpha_{11}}, \ldots, \frac{\partial}{\partial \alpha_{1p}} \right) \left(\alpha_{11}^2 + \cdots + \alpha_{1p}^2 \right) \\
&= (2\alpha_{11}, \ldots, 2\alpha_{1p}) \\
&= 2\boldsymbol{\alpha}_1
\end{aligned}$$

(And that's a general result: $\nabla \|\boldsymbol{x}\|_2^2 = 2\boldsymbol{x}$ with $\|\cdot\|_2$ the Euclidean norm)

Effect of ∇ on f

Expand (write $\Sigma = [s_{ij}]$ and do not exploit symmetry)

$$\boldsymbol{\alpha}_1^T \Sigma \boldsymbol{\alpha}_1 = (\alpha_{11}, \ldots, \alpha_{1p}) \begin{pmatrix} s_{11} & s_{12} & \cdots & s_{1p} \\ s_{21} & s_{22} & \cdots & s_{2p} \\ \vdots & \vdots & & \vdots \\ s_{p1} & s_{p2} & & s_{pp} \end{pmatrix} \begin{pmatrix} \alpha_{11} \\ \alpha_{12} \\ \vdots \\ \alpha_{1p} \end{pmatrix}$$

$$= (\alpha_{11}, \ldots, \alpha_{1p}) \begin{pmatrix} s_{11}\alpha_{11} + s_{12}\alpha_{12} + \cdots + s_{1p}\alpha_{1p} \\ s_{21}\alpha_{11} + s_{22}\alpha_{12} + \cdots + s_{2p}\alpha_{1p} \\ \vdots \\ s_{p1}\alpha_{11} + s_{p2}\alpha_{12} + \cdots + s_{pp}\alpha_{1p} \end{pmatrix}$$

$$= (s_{11}\alpha_{11} + s_{12}\alpha_{12} + \cdots + s_{1p}\alpha_{1p})\alpha_{11}$$
$$+ (s_{21}\alpha_{11} + s_{22}\alpha_{12} + \cdots + s_{2p}\alpha_{1p})\alpha_{12}$$
$$\vdots$$
$$+ (s_{p1}\alpha_{11} + s_{p2}\alpha_{12} + \cdots + s_{pp}\alpha_{1p})\alpha_{1p}$$

We have

$$\begin{aligned}\boldsymbol{\alpha}_1^T \Sigma \boldsymbol{\alpha}_1 = &(s_{11}\alpha_{11} + s_{12}\alpha_{12} + \cdots + s_{1p}\alpha_{1p})\alpha_{11} \\ &+ (s_{21}\alpha_{11} + s_{22}\alpha_{12} + \cdots + s_{2p}\alpha_{1p})\alpha_{12} \\ &\vdots \\ &+ (s_{p1}\alpha_{11} + s_{p2}\alpha_{12} + \cdots + s_{pp}\alpha_{1p})\alpha_{1p}\end{aligned}$$

$$\begin{aligned}\implies \frac{\partial}{\partial \alpha_{11}} \boldsymbol{\alpha}_1^T \Sigma \boldsymbol{\alpha}_1 &= (s_{11}\alpha_{11} + s_{12}\alpha_{12} + \cdots + s_{1p}\alpha_{1p}) + s_{11}\alpha_{11} \\ &\quad + s_{21}\alpha_{12} + \cdots + s_{p1}\alpha_{1p} \\ &= s_{11}\alpha_{11} + s_{12}\alpha_{12} + \cdots + s_{1p}\alpha_{1p} \\ &\quad + s_{11}\alpha_{11} + s_{21}\alpha_{12} + \cdots + s_{p1}\alpha_{1p} \\ &= 2(s_{11}\alpha_{11} + s_{12}\alpha_{12} + \cdots + s_{1p}\alpha_{1p})\end{aligned}$$

(last equality stems from symmetry of Σ)

In general, for $i = 1, \ldots, p$,

$$\begin{aligned}\frac{\partial}{\partial \alpha_{1i}} \boldsymbol{\alpha}_1^T \Sigma \boldsymbol{\alpha}_1 &= s_{i1}\alpha_{11} + s_{i2}\alpha_{12} + \cdots + s_{ip}\alpha_{1p} \\ &\quad + s_{i1}\alpha_{11} + s_{2i}\alpha_{12} + \cdots + s_{pi}\alpha_{1p} \\ &= 2(s_{i1}\alpha_{11} + s_{i2}\alpha_{12} + \cdots + s_{ip}\alpha_{1p})\end{aligned}$$

(because of symmetry of Σ)

As a consequence,

$$\nabla \boldsymbol{\alpha}_1^T \Sigma \boldsymbol{\alpha}_1 = 2\Sigma \boldsymbol{\alpha}_1$$

So solving

$$\nabla f(x_1, \ldots, x_n) = \lambda \nabla g(x_1, \ldots, x_n)$$

means solving

$$2\Sigma\boldsymbol{\alpha}_1 = \lambda 2\boldsymbol{\alpha}_1$$

i.e.,

$$\Sigma\boldsymbol{\alpha}_1 = \lambda\boldsymbol{\alpha}_1$$

$\implies$ $(\lambda, \boldsymbol{\alpha}_1)$ eigenpair of Σ, with $\boldsymbol{\alpha}_1$ having unit length

Picking the right eigenvalue

$(\lambda, \boldsymbol{\alpha}_1)$ eigenpair of Σ, with $\boldsymbol{\alpha}_1$ having unit length

But which λ to choose?

Recall that we want Var $\boldsymbol{\alpha}_1^T \boldsymbol{x} = \boldsymbol{\alpha}_1^T \Sigma \boldsymbol{\alpha}_1$ maximal

We have

$$\text{Var } \boldsymbol{\alpha}_1^T \boldsymbol{x} = \boldsymbol{\alpha}_1^T \Sigma \boldsymbol{\alpha}_1 = \boldsymbol{\alpha}_1^T (\Sigma \boldsymbol{\alpha}_1) = \boldsymbol{\alpha}_1^T (\lambda \boldsymbol{\alpha}_1) = \lambda (\boldsymbol{\alpha}_1^T \boldsymbol{\alpha}_1) = \lambda$$

$\Longrightarrow$ we pick $\lambda = \lambda_1$, the largest eigenvalue (covariance matrix symmetric so eigenvalues real)

What we have this far..

The first principal component is $\boldsymbol{\alpha}_1^T \boldsymbol{x}$ and has variance λ_1, where λ_1 the largest eigenvalue of Σ and $\boldsymbol{\alpha}_1$ an associated eigenvector with $\|\boldsymbol{\alpha}_1\| = 1$

We want the second principal component to be *uncorrelated* with $\boldsymbol{\alpha}_1^T \boldsymbol{x}$ and to have maximum variance Var $\boldsymbol{\alpha}_2^T \boldsymbol{x} = \boldsymbol{\alpha}_2^T \Sigma \boldsymbol{\alpha}_2$, under the constraint that $\|\boldsymbol{\alpha}_2\| = 1$

$\boldsymbol{\alpha}_2^T \boldsymbol{x}$ uncorrelated to $\boldsymbol{\alpha}_1^T \boldsymbol{x}$ if $\text{cov}(\boldsymbol{\alpha}_1^T \boldsymbol{x}, \boldsymbol{\alpha}_2^T \boldsymbol{x}) = 0$

We have

$$\begin{aligned}
\text{cov}(\boldsymbol{\alpha}_1^T \boldsymbol{x}, \boldsymbol{\alpha}_2^T \boldsymbol{x}) &= \boldsymbol{\alpha}_1^T \Sigma \boldsymbol{\alpha}_2 \\
&= \boldsymbol{\alpha}_2^T \Sigma^T \boldsymbol{\alpha}_1 \\
&= \boldsymbol{\alpha}_2^T \Sigma \boldsymbol{\alpha}_1 \quad [\Sigma \text{ symmetric}] \\
&= \boldsymbol{\alpha}_2^T (\lambda_1 \boldsymbol{\alpha}_1) \\
&= \lambda \boldsymbol{\alpha}_2^T \boldsymbol{\alpha}_1
\end{aligned}$$

So $\boldsymbol{\alpha}_2^T \boldsymbol{x}$ uncorrelated to $\boldsymbol{\alpha}_1^T \boldsymbol{x}$ if $\boldsymbol{\alpha}_1 \perp \boldsymbol{\alpha}_2$

This is beginning to sound a lot like Gram-Schmidt, no?

In short

Take whatever covariance matrix is available to you (known Σ or sample S_X) – assume sample from now on for simplicity

For $i = 1, \ldots, p$, the ith principal component is

$$z_i = \boldsymbol{v}_i^T \boldsymbol{x}$$

where $\boldsymbol{v}_i$ eigenvector of S_X associated to the ith largest eigenvalue λ_i

If $\boldsymbol{v}_i$ is normalised, then $\lambda_i = \text{Var } z_i$

A 3D example – Sample covariance matrix

Suppose we have a dataset with $p = 3$ variables, $\boldsymbol{x} = (x_1, x_2, x_3)^T$, centred so that sample means are zero

After computing all sample variances and covariances, we obtain the sample covariance matrix:

$$S_X = \begin{pmatrix} 15 & 10 & 4 \\ 10 & 14 & 6 \\ 4 & 6 & 7 \end{pmatrix}$$

- Sample variances (diagonal): Var $(x_1) = 15$, Var $(x_2) = 14$, Var $(x_3) = 7$
- Sample covariances (off-diagonal): all variables are positively correlated ($\text{cov}(x_1, x_2) = 10$, $\text{cov}(x_1, x_3) = 4$, $\text{cov}(x_2, x_3) = 6$)
- Total sample variance in the dataset is the trace:

$$\text{tr}(S_X) = 15 + 14 + 7 = 36$$

A 3D example – Eigenvalues

To compute the principal components, we find the eigenvalues of S_X by solving $\det(S_X - \lambda I) = 0$:

$$\det \begin{pmatrix} 15-\lambda & 10 & 4 \\ 10 & 14-\lambda & 6 \\ 4 & 6 & 7-\lambda \end{pmatrix} = 0$$

This yields three distinct positive eigenvalues:

$$\lambda_1 = 27, \qquad \lambda_2 = 6, \qquad \lambda_3 = 3$$

Note that:

$$\lambda_1 + \lambda_2 + \lambda_3 = 27 + 6 + 3 = 36 = \mathrm{tr}(S_X)$$

As expected from theory, the total variance is preserved!

A 3D example – Variance explained

The variance of each principal component $z_i = \boldsymbol{v}_i^T \boldsymbol{x}$ is given by Var $z_i = \lambda_i$ (for normalized $\boldsymbol{v}_i$)

Proportion of total variance explained by each principal component:

- **1st PC (z_1)**:
$$\frac{\lambda_1}{\text{tr}(S_X)} = \frac{27}{36} = \frac{3}{4} = 75\%$$
- **2nd PC (z_2)**:
$$\frac{\lambda_2}{\text{tr}(S_X)} = \frac{6}{36} = \frac{1}{6} \approx 16.67\%$$
- **3rd PC (z_3)**:
$$\frac{\lambda_3}{\text{tr}(S_X)} = \frac{3}{36} = \frac{1}{12} \approx 8.33\%$$

The first two principal components together account for $\frac{27+6}{36} = \frac{33}{36} \approx 91.67\%$ of the total variance

A 3D example – Eigenvectors

Next, we find the associated normalized eigenvectors by solving $(S_X - \lambda_i I)\boldsymbol{v}_i = \mathbf{0}$ subject to $\|\boldsymbol{v}_i\| = 1$:

- For $\lambda_1 = 27$:

$$\boldsymbol{v}_1 = \frac{1}{3}\begin{pmatrix}2\\2\\1\end{pmatrix} = \begin{pmatrix}2/3\\2/3\\1/3\end{pmatrix}, \qquad \|\boldsymbol{v}_1\|^2 = \frac{4+4+1}{9} = 1$$

- For $\lambda_2 = 6$:

$$\boldsymbol{v}_2 = \frac{1}{3}\begin{pmatrix}-2\\1\\2\end{pmatrix} = \begin{pmatrix}-2/3\\1/3\\2/3\end{pmatrix}, \qquad \|\boldsymbol{v}_2\|^2 = \frac{4+1+4}{9} = 1$$

- For $\lambda_3 = 3$:

$$\boldsymbol{v}_3 = \frac{1}{3}\begin{pmatrix}1\\-2\\2\end{pmatrix} = \begin{pmatrix}1/3\\-2/3\\2/3\end{pmatrix}, \qquad \|\boldsymbol{v}_3\|^2 = \frac{1+4+4}{9} = 1$$

A 3D example – Orthonormality check

Since S_X is symmetric with distinct eigenvalues, the eigenvectors are mutually orthogonal:

$$\boldsymbol{v}_1^T \boldsymbol{v}_2 = \frac{1}{9}\big(2(-2) + 2(1) + 1(2)\big) = \frac{-4+2+2}{9} = 0$$
$$\boldsymbol{v}_1^T \boldsymbol{v}_3 = \frac{1}{9}\big(2(1) + 2(-2) + 1(2)\big) = \frac{2-4+2}{9} = 0$$
$$\boldsymbol{v}_2^T \boldsymbol{v}_3 = \frac{1}{9}\big(-2(1) + 1(-2) + 2(2)\big) = \frac{-2-2+4}{9} = 0$$

Hence, $V = \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \boldsymbol{v}_3 \end{pmatrix}$ is an orthogonal matrix ($V^T V = I$), forming an orthonormal basis of $\mathbb{R}^3$

A 3D example – The principal components

The principal components are the linear combinations $z_i = \boldsymbol{v}_i^T \boldsymbol{x}$:

$$z_1 = \boldsymbol{v}_1^T \boldsymbol{x} = \frac{2}{3}x_1 + \frac{2}{3}x_2 + \frac{1}{3}x_3$$
$$z_2 = \boldsymbol{v}_2^T \boldsymbol{x} = -\frac{2}{3}x_1 + \frac{1}{3}x_2 + \frac{2}{3}x_3$$
$$z_3 = \boldsymbol{v}_3^T \boldsymbol{x} = \frac{1}{3}x_1 - \frac{2}{3}x_2 + \frac{2}{3}x_3$$

Properties of these new variables:

- Var $z_1 = \lambda_1 = 27$, Var $z_2 = \lambda_2 = 6$, Var $z_3 = \lambda_3 = 3$
- $\text{cov}(z_i, z_j) = 0$ for all $i \neq j$ (uncorrelated)
- In the new basis, the covariance matrix is diagonal:

$$S_Z = V^T S_X V = \text{diag}(27, 6, 3)$$

A 3D example – Dimensionality reduction

Suppose we want to reduce the dimension of the data from 3 to 2:

- Keep only the first two components, z_1 and z_2
- z_1 and z_2 together preserve 91.67% of the total variance
- Each 3D data point $\boldsymbol{x} = (x_1, x_2, x_3)^T$ is mapped to its 2D coordinates $(z_1, z_2)^T$
- Approximation of $\boldsymbol{x}$ using only 2 components:

$$\boldsymbol{x} \approx z_1 \boldsymbol{v}_1 + z_2 \boldsymbol{v}_2$$

- Discarding z_3 loses only 8.33% of the variance

A 3D example – Verifying in R

We can confirm our calculations directly in R:

```
S = matrix(c(15, 10, 4, 10, 14, 6, 4, 6, 7), nrow = 3, byrow = TRUE)
ev = eigen(S)
ev$values
```

```
## [1] 27  6  3
```

```
# Proportion and cumulative proportion of variance explained
ev$values / sum(ev$values)
```

```
## [1] 0.75000000 0.16666667 0.08333333
```

```
cumsum(ev$values) / sum(ev$values)
```

```
## [1] 0.7500000 0.9166667 1.0000000
```

Covariance

The function `cov` returns the covariance of two samples

Note that the functions deals equally well with data that is not centred as with data that is centred

```
cov(data$height, data$weight)

## [1] 25.98646

cov(data$height.c, data$weight.c)

## [1] 25.98646
```

Covariance matrix

As we could see from plotting the data, there is a positive linear relationship between the two variables

Let us compute the sample covariance matrix

```
X = as.matrix(data[,c("height.c", "weight.c")])
S = 1/(dim(X)[1]-1)*t(X) %*% X
S

##           height.c weight.c
## height.c 28.98592 25.98646
## weight.c 25.98646 48.49557
```

Covariance matrix

The off-diagonal entries do match the computed covariance. Let us check that the variances are indeed a match too.

```
var(X[,1])

## [1] 28.98592

var(X[,2])

## [1] 48.49557
```

Hey, that works. Is math not cool? ;)

Principal components

Now compute the principal components. We need eigenvalues and eigenvectors

```
ev = eigen(S)
ev

## eigen() decomposition
## $values
## [1] 66.49777 10.98372
##
## $vectors
##            [,1]       [,2]
## [1,] 0.5694576 -0.8220207
## [2,] 0.8220207  0.5694576
```

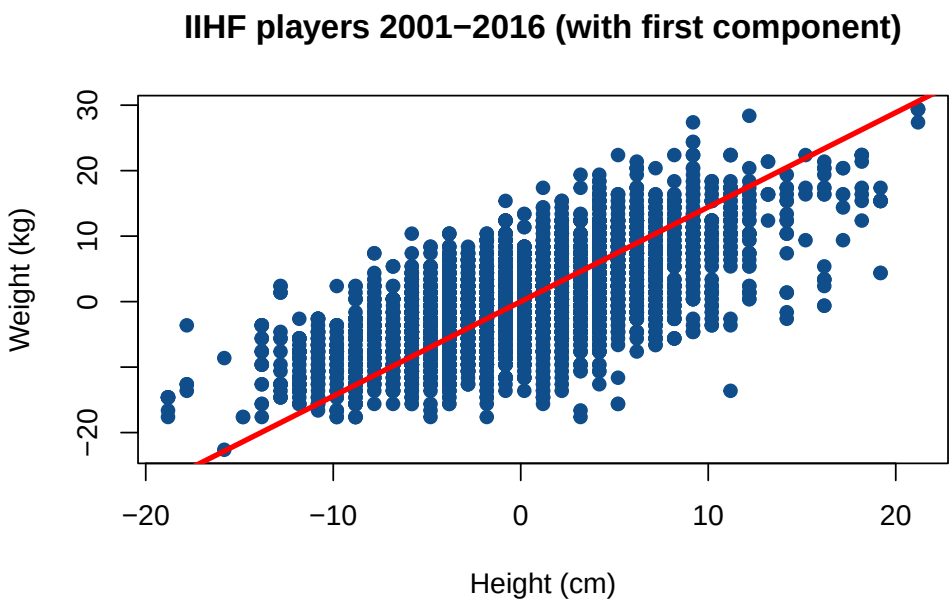
(`eigen` returns eigenvalues sorted in decreasing order and normalised eigenvectors)

First principal component

Let us plot this first eigenvector (well, the line carrying this first eigenvector)

To use the function `abline`, we need to give the coefficients of the line in the form of (intercept,slope). Intercept is easy, as the line goes through the origin (by construction and because we have centred the data). The slope is also quite simple..

```
plot(data$height.c, data$weight.c,
    pch = 19, col = "dodgerblue4",
    main = "IIHF players 2001-2016 (with first component)",
    xlab = "Height (cm)", ylab = "Weight (kg)")
abline(a = 0, b = ev$vectors[2,1]/ev$vectors[1,1],
       col = "red", lwd = 3)
```


IIHF players 2001−2016 (with first component)
Weight (kg)
−20
0
10
20
30
−20
−10
0
10
20
Height (cm)

Rotating the data

Let us rotate the data so that the red line becomes the x-axis

To do that, we use a rotation matrix

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

To find the angle θ, recall that $\tan\theta$ is equal to opposite length over adjacent length, i.e.,

$$\tan\theta = \frac{\texttt{ev\$vectors}[2,1]}{\texttt{ev\$vectors}[1,1]}$$

So we just use the $\arctan$ of this

Note that angles are in radians

Rotating the data

```
theta = atan(ev$vectors[2,1]/ev$vectors[1,1])
theta
```

```
## [1] 0.9649505
```

```
R_theta = matrix(c(cos(theta), -sin(theta),
                   sin(theta), cos(theta)),
                 nr = 2, byrow = TRUE)
R_theta
```

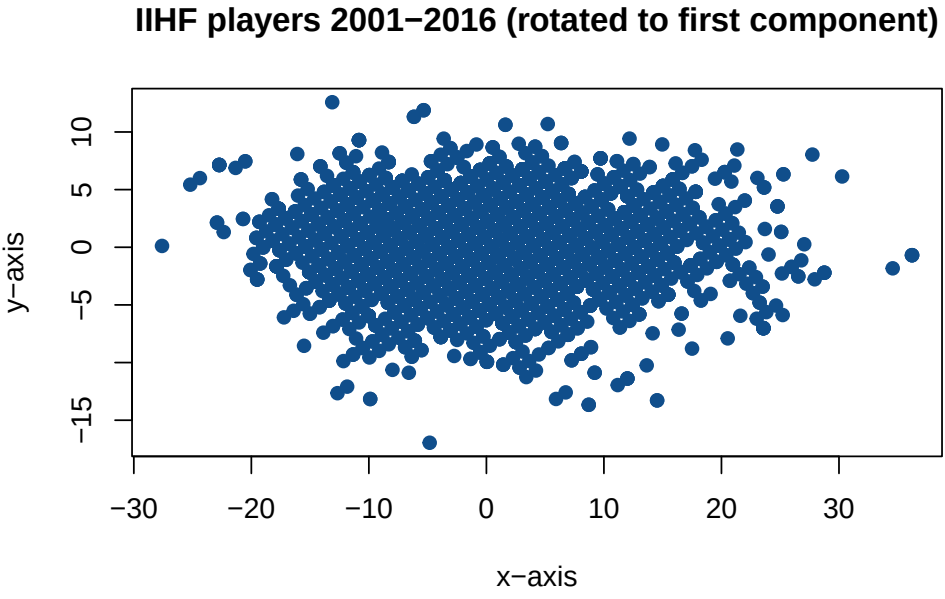
```
##           [,1]       [,2]
## [1,] 0.5694576 -0.8220207
## [2,] 0.8220207  0.5694576
```

Rotating the data

And now we rotate the points

(In this case, we think of the points as vectors, of course)

```
tmp_in = matrix(c(data$weight.c, data$height.c),
                nc = 2)
tmp_out = c()
for (i in 1:dim(tmp_in)[1]) {
    tmp_out = rbind(tmp_out,
                    t(R_theta %*% tmp_in[i,]))
}
data$weight.c_r = tmp_out[,1]
data$height.c_r = tmp_out[,2]
```

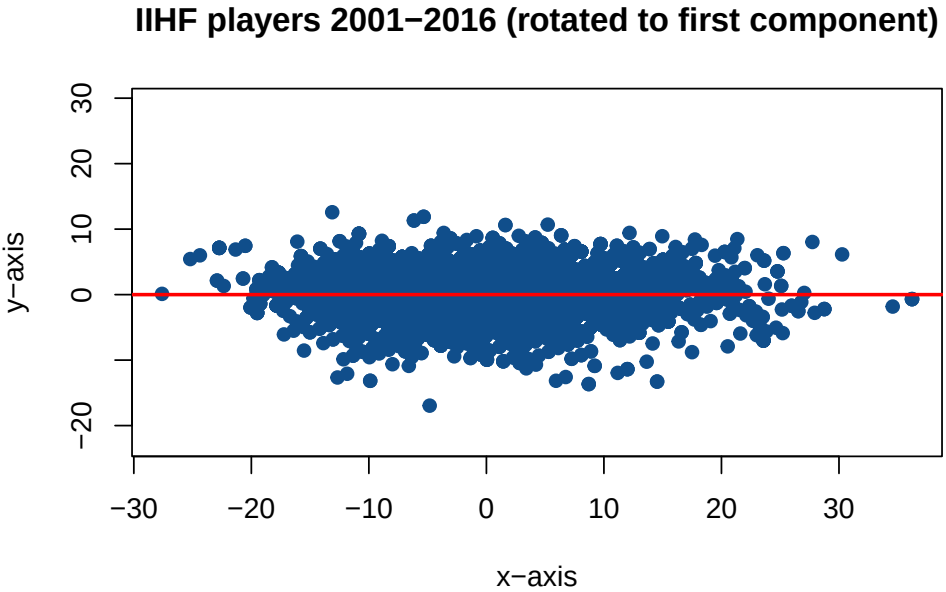

IIHF players 2001−2016 (rotated to first component)
y−axis
10
5
0
−5
−15
−30
−20
−10
0
10
20
30
x−axis

Principal components

Note that the axes have changed quite a lot, hence the very different aspect

Let us plot with the same range as for the non-rotated data for the y-axis

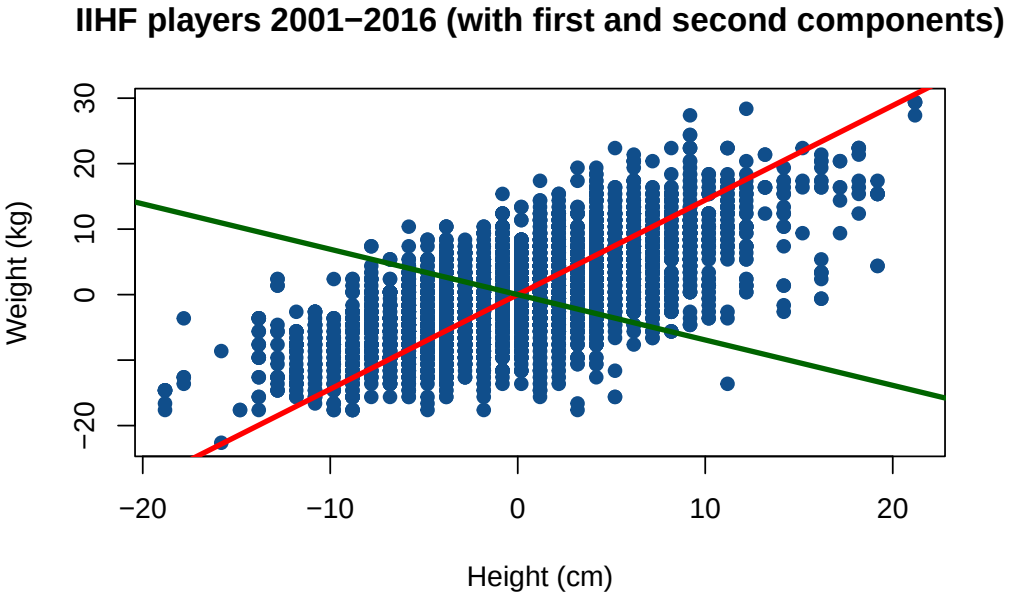
```
plot(data$height.c_r, data$weight.c_r,
    pch = 19, col = "dodgerblue4",
    xlab = "x-axis", ylab = "y-axis",
    main = "IIHF players 2001-2016 (rotated to first component)",
    ylim = range(data$weight.c))
abline(h = 0, col = "red", lwd = 2)
```


IIHF players 2001−2016 (rotated to first component)
y−axis
30
20
10
0
−20
−30
−20
−10
0
10
20
30
x−axis

First and second principal components

Plot the first and second eigenvectors

```
plot(data$height.c, data$weight.c,
    pch = 19, col = "dodgerblue4",
    main = "IIHF players 2001-2016 (with first and second components)",
    xlab = "Height (cm)", ylab = "Weight (kg)")
abline(a = 0, b = ev$vectors[2,1]/ev$vectors[1,1],
       col = "red", lwd = 3)
abline(a = 0, b = ev$vectors[2,2]/ev$vectors[1,2],
       col = "darkgreen", lwd = 3)
```


IIHF players 2001−2016 (with first and second components)
Weight (kg)
30
20
10
0
−20
−20
−10
0
10
20
Height (cm)

Proper change of basis

Let us change the basis so that, in the new basis, the first component is the x-axis and the second component is the y-axis

We want to use Theorem **??**

We need the coordinates of the new basis in the canonical basis of $\mathbb{R}^2$

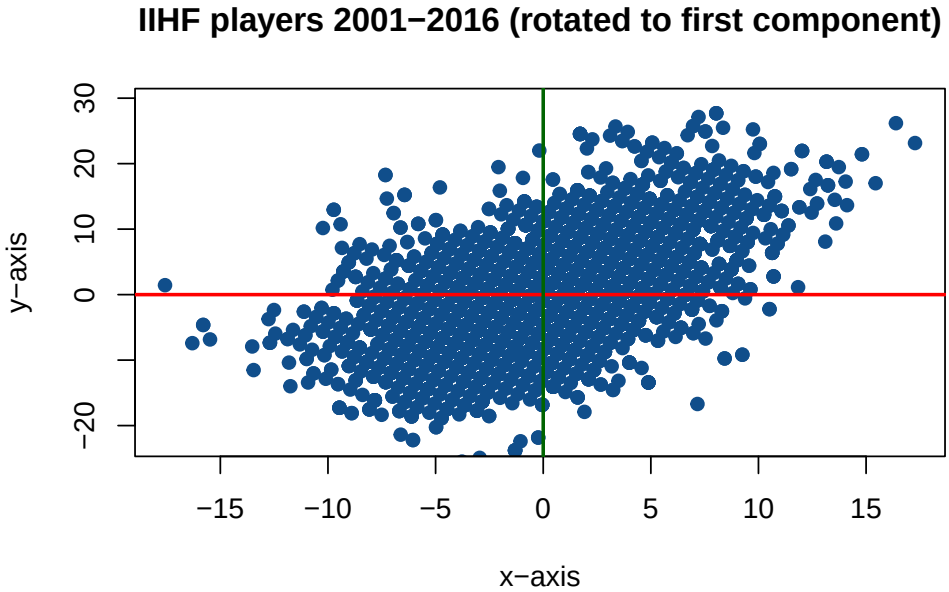
Since both axes go through the origin, we can just use $y = ax$, with a the slope of the lines and, say, $x = 1$, i.e., $(x, y) = (1, a)$

We then normalise the resulting vectors

Proper change of basis

```
red_line = c(1, ev$vectors[2,1]/ev$vectors[1,1])
red_line = red_line/sqrt(sum(red_line^2))
green_line = c(1, ev$vectors[2,2]/ev$vectors[1,2])
green_line = green_line/sqrt(sum(green_line^2))
augmented_M = cbind(red_line,green_line, diag(2))
P = rref(augmented_M)[,3:4]

tmp_in = matrix(c(data$weight.c, data$height.c), nc = 2)
tmp_out = c()
for (i in 1:dim(tmp_in)[1]) {
    tmp_out = rbind(tmp_out, t(P %*% tmp_in[i,]))
}
data$weight.c_r2 = tmp_out[,1]
data$height.c_r2 = tmp_out[,2]
```


IIHF players 2001−2016 (rotated to first component)
y−axis
30
20
10
0
−20
−15
−10
−5
0
5
10
15
x−axis

PCA using built-in functions

Now do things “properly”

```
GS = pracma::gramSchmidt(A = ev$vectors, tol = 1e-10)
GS

## $Q
##           [,1]       [,2]
## [1,] 0.5694576 -0.8220207
## [2,] 0.8220207  0.5694576
##
## $R
##      [,1]          [,2]
## [1,]    1 -5.551115e-17
## [2,]    0  1.000000e+00
```

PCA using built-in functions

Now recall we saw a theorem that told us how to construct a new basis..

```
A=matrix(c(GS$Q,1,0,0,1), nr = 2)
A
```

```
##            [,1]       [,2] [,3] [,4]
## [1,] 0.5694576 -0.8220207    1    0
## [2,] 0.8220207  0.5694576    0    1
```

```
pracma::rref(A)
```

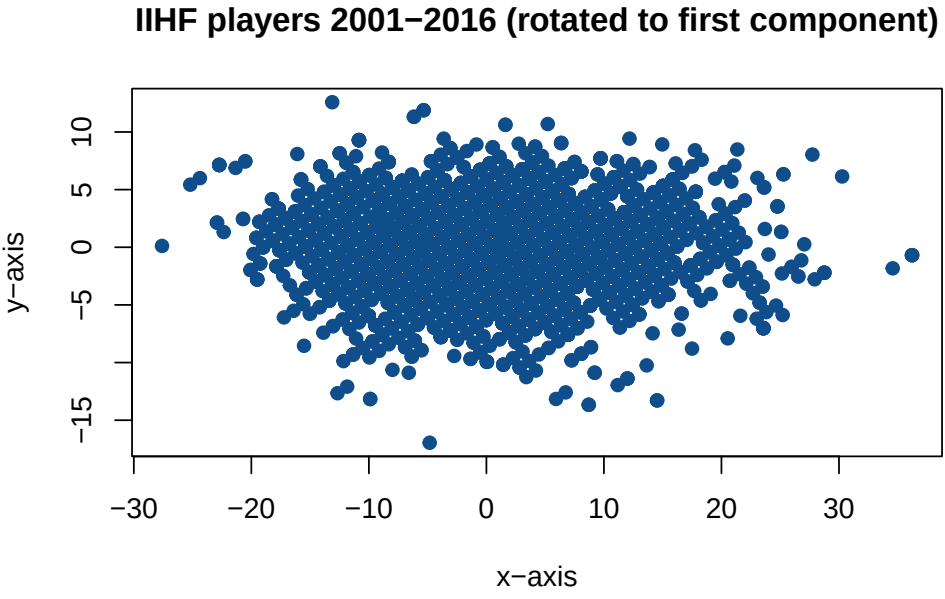
```
##      [,1] [,2]       [,3]      [,4]
## [1,]    1    0  0.5694576 0.8220207
## [2,]    0    1 -0.8220207 0.5694576
```

PCA using built-in functions

```
P = pracma::rref(A)[,c(3,4)]

##             [,1]       [,2]
## [1,]  0.5694576 0.8220207
## [2,] -0.8220207 0.5694576

X.new = X %*% t(P)
```


IIHF players 2001−2016 (rotated to first component)
y−axis
10
5
0
−5
−15
−30
−20
−10
0
10
20
30
x−axis